BIKE BUYERS CUSTOMER ANALYSIS

Customer Segmentation, Purchase Conversion and Regional Performance

Tool: Microsoft Excel
Dataset: 1,000 Customer Records
Analysis Type: Exploratory Data Analysis and Customer Segmentation
Deliverable: Interactive Excel Dashboard

1. Executive Summary

An analysis of 1,000 customer records revealed that bicycle purchasing behavior varied more noticeably across age groups, commuting distances, and geographic regions than across gender.

Of the 1,000 customers analyzed, 481 purchased bicycles, representing an overall purchase rate of 48.1%. Customers aged 30 to 39 recorded the highest age-group purchase rate at 59.5%, while those commuting 2 to 5 miles achieved a 58.6% purchase rate.

Regional performance also revealed an important commercial opportunity. Although the Pacific region achieved the highest purchase rate at 58.9%, North America accounted for the largest customer population, with 508 customers, but recorded the lowest regional purchase rate at 43.3%.

These findings suggest two potential business priorities: testing acquisition campaigns among customer segments with relatively high purchase rates and investigating opportunities to improve conversion within North America's larger customer base.

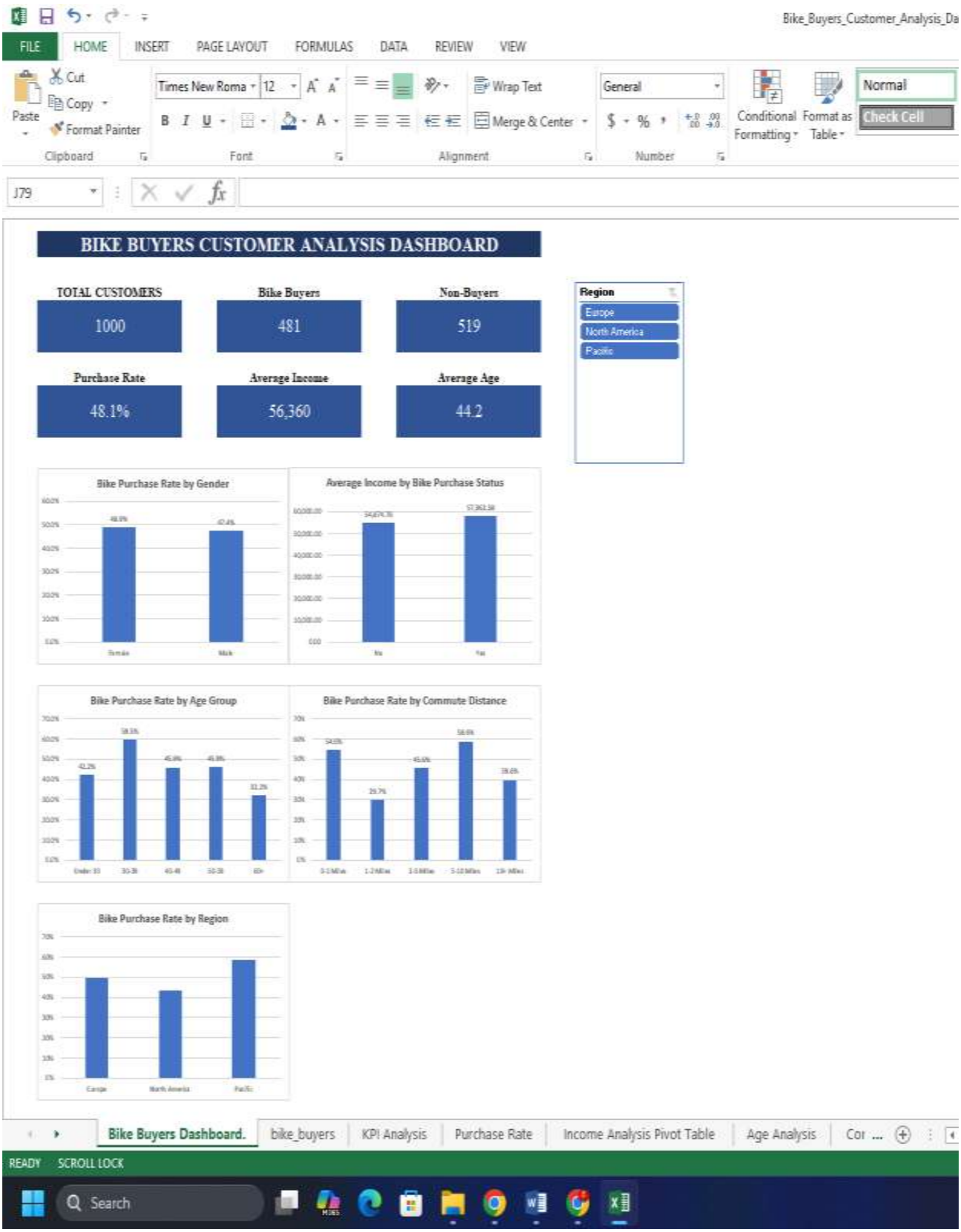


Figure 1: Bike Buyers Customer Analysis Dashboard showing customer KPIs, purchasing patterns, and interactive regional filtering.

2. Business Problem and Analytical Objectives

The central business question was:

Which customer characteristics are associated with bicycle purchases, and where could a retailer focus its marketing and conversion improvement efforts?

To answer this question, the analysis examined five areas:

1. How purchase rates differ across customer age groups.
2. How commuting distance relates to bicycle purchasing.
3. Which regions demonstrate stronger or weaker purchase conversion.
4. How average income differs between buyers and non-buyers.
5. How purchasing behavior compares between male and female customers.

The objective was not simply to identify the segment with the most purchases, but to distinguish customer volume from purchase rate and use both measures to assess commercial opportunities.

3. Data Preparation and Quality Checks

The original dataset contained 1,026 records across 13 variables, including customer ID, income, gender, age, commute distance, region, and purchase status.

During data cleaning, 26 duplicate customer records were identified and removed after confirming that the duplicate entries were identical. This produced a final analytical dataset of 1,000 unique records.

The dataset was also checked for missing values and appropriate numeric data types. No missing values were identified. A new Age Group field was created to divide customers into five categories: Under 30, 30 to 39, 40 to 49, 50 to 59, and 60 or older.

This preparation ensured that customer counts were not inflated by duplicate records and that age-based comparisons could be performed consistently.

4. Dashboard Development and Analytical Method

The analysis was performed in Microsoft Excel using Pivot Tables, conditional formulas, calculated purchase rates, charts, and slicers.

Six key performance indicators were developed:

KPI	Overall Result
Total Customers	1,000
Bike Buyers	481

Non-Buyers	519
Purchase Rate	48.1%
Average Customer Income	56,360
Average Customer Age	44.2 years

Five charts were created to compare purchasing patterns by gender, income, age group, commuting distance, and region.

A Region slicer was connected to the relevant Pivot Tables, charts, and KPI calculations, allowing users to filter the dashboard by Europe, North America, or Pacific.

The interactive calculations were validated against known regional customer counts and purchase rates to confirm that filtering produced consistent results.

Dashboard Screenshot: Insert a screenshot of the completed Excel dashboard here.

5. Key Analytical Findings

Finding 1: Customers Aged 30 to 39 Had the Highest Purchase Rate

The 30 to 39 age group recorded 179 purchases among 301 customers, producing a purchase rate of 59.5%.

In comparison, customers aged 60 or older recorded 37 purchases among 115 customers, representing a purchase rate of 32.2%.

The difference between these two age groups was approximately 27.3 percentage points.

This indicates that age-based purchasing patterns differ substantially within the dataset. However, the results do not establish that age itself causes a customer to purchase a bicycle.

Finding 2: Moderate Commuting Distances Were Associated With Higher Purchase Rates

Customers commuting 2 to 5 miles recorded the highest purchase rate at 58.6%, followed by those commuting 0 to 1 mile at 54.6%.

Together, customers commuting 0 to 5 miles accounted for 295 purchases among 697 customers, representing approximately 61.3% of all bicycle purchases.

By comparison, customers commuting 10 miles or more recorded a purchase rate of just 29.7%.

This pattern suggests that shorter and moderate commuting distances may be useful variables when identifying audiences for commuter-oriented bicycle products.

Finding 3: North America Presented a Conversion Improvement Opportunity

Regional analysis revealed the following results:

Region	Customers	Bike Buyers	Purchase Rate
Europe	300	148	49.3%
North America	508	220	43.3%
Pacific	192	113	58.9%

The Pacific region achieved the highest purchase rate, but North America generated the largest absolute number of purchases.

North America also had the lowest conversion rate, creating a potentially important improvement opportunity.

For illustration, increasing North America's purchase rate by five percentage points, from approximately 43.3% to 48.3%, would correspond to about 25 additional purchases across a comparable group of 508 customers.

This is a scenario calculation, not a forecast. Actual results would depend on the effectiveness of any intervention.

Finding 4: Buyers Had Higher Average Incomes Than Non-Buyers

Customers who purchased bicycles had an average income of approximately 57,963, compared with 54,875 among non-buyers.

The difference of approximately 3,088 represents a 5.6% higher average income among buyers.

This suggests a possible association between income and bicycle purchasing. However, the difference alone does not establish purchasing power as the primary explanation. Further analysis by income band, region, and age would help determine how consistent this relationship is.

Finding 5: Gender Showed a Relatively Small Difference in Purchase Rates

Female customers recorded a purchase rate of 48.9%, compared with 47.4% among male customers.

The difference of approximately 1.5 percentage points was much smaller than the observed differences between certain age groups, commuting categories, and regions.

Based on these descriptive results, gender appears less useful as a primary customer segmentation variable than age, commuting distance, or geographic location.

6. Strategic Business Recommendations

Recommendation 1: Test Campaigns for Customers Aged 30 to 39

This group recorded 179 purchases and the highest age-group purchase rate at 59.5%. Test commuter-focused campaigns for this audience and compare conversion rates and acquisition costs with other age segments.

Recommendation 2: Develop Short-Distance Commuter Offers

Customers commuting 0 to 5 miles accounted for 295 bicycle purchases. Test bicycle bundles featuring helmets, locks, and maintenance services, then evaluate average order value and profitability.

Recommendation 3: Investigate North America's Lower Conversion Rate

North America represents 50.8% of the customer dataset but recorded the lowest regional purchase rate at 43.3%. Investigate pricing, product suitability, and customer objections before testing targeted conversion improvements.

Recommendation 4: Evaluate Product Offers Across Income Bands

Buyers earned approximately 5.6% more than non-buyers. Test value-oriented and premium bicycle propositions across income segments, measuring conversion and profit margins rather than assuming higher income guarantees a purchase.

Recommendation 5: Prioritize Behavioral Segmentation

Gender purchase rates differed by only 1.5 percentage points. Give greater priority to testing segmentation strategies based on age, commute distance, and region, where the observed differences were larger.

7. Limitations of the Analysis

The analysis is descriptive and does not establish causal relationships.

The dataset does not include transaction dates, product prices, marketing campaign exposure, customer acquisition costs, or profit margins. As a result, the analysis cannot determine seasonal purchasing trends, marketing return on investment, or the financial impact of the proposed recommendations.

Additionally, the observed purchase rates describe this dataset and should not automatically be assumed to represent the retailer's entire customer market.

These limitations should be considered before using the findings to make major commercial investments.

8. Project Outcome and Business Value

The project produced a validated, interactive Excel dashboard that enables users to compare customer purchasing patterns across three regions and monitor changes in six performance indicators.

The analysis identified high-purchase-rate segments, highlighted a potential conversion improvement opportunity in North America, and provided measurable recommendations for further business testing.

The dashboard offers a practical tool for exploratory customer analysis and a foundation for more advanced segmentation, campaign measurement, and sales performance evaluation.

9. Technical Skills Demonstrated

Microsoft Excel, Data Cleaning, Duplicate Detection, Data Validation, IF Formulas, COUNTIF, AVERAGE, Pivot Tables, GETPIVOTDATA, KPI Calculations, Interactive Slicers, Dashboard Development, Customer Segmentation, Data Visualization, Business Analysis, and Analytical Reporting.